

# Growth Degree of the Mittag-Leffler Function of One Parameter

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## 1 Introduction

The Mittag-Leffler Function is a generalization of the exponential function  $e^x$  to fractional calculus. It connects the regular calculus, where integer degrees of differentiation and integration is studied, to fractional calculus where fractional degrees of differentiation and integration is examined. It is used extensively in fractional calculus and Fractional integral/differential equations. It has applications on physics, varying from Quantum Field Theory and Non-Markovian processes to Nuclear Physics. The Mittag-Leffler function of one parameter can be generalized as below:

$$E_\alpha(z) = \sum_{n=0}^{\infty} \frac{z^n}{\Gamma(n\alpha + 1)}$$

By adding a parameter alpha into the input of gamma function, we expand the domain of the indicator n from non-negative integers to non-negative rational numbers. By doing so, we can control the growth degree( $\rho$ ) of the function we obtain.

## 2 Evaluation of Growth Degree via the Maximum Modulus

Lets apply the generalized limit definition for growth degree. The growth degree of some entire function (analytic on the whole  $\mathbb{C}$ )  $f(z)$  is given by:

$$\rho(f(z)) = \limsup_{r \rightarrow \infty} \left( \frac{\ln(\ln(M(r)))}{\ln(r)} \right), \quad M(r) = \max_{r=|z|} |f(z)|$$

where  $M(r)$  is the **maximum modulus** of  $f(z)$  on a circle of radius  $r = |z|$ . The direction that maximizes the modulus of  $E_\alpha(z)$  where  $r \rightarrow \infty$  is the behavior along  $r = z$  or the positive real part of the complex plane.

According to Wright and Djrbashian expansions, the main asymptotic term of the function  $E_\alpha(z)$  where  $r \rightarrow \infty$  is given as below:

$$E_\alpha(r) \sim \frac{1}{\alpha} \exp\left(r^{1/\alpha}\right)$$

Therefore, asymptotic behavior of maximum modulus for big values of  $r$  is given in the form:

$$M(r) \sim \frac{1}{\alpha} e^{r^{1/\alpha}}, \text{ where } (r \rightarrow \infty)$$

By the definition of similarities, this expression can be written in the following form:

$$p(x) \sim q(x) \implies \lim_{x \rightarrow \infty} \frac{p(x)}{q(x)} = 1$$

Because that they are not equal but similar, we have to add an asymptotic correction term  $\epsilon(r)$  to the expression when forming an equality. The small error between them will approach to zero as  $r$  approaches to infinity.

$$M(r) \sim \frac{1}{\alpha} e^{r^{1/\alpha}} \therefore M(r) = \frac{1}{\alpha} e^{r^{1/\alpha}} \cdot (1 + \epsilon(r)), \text{ where } \lim_{r \rightarrow \infty} \epsilon(r) = 0$$

If we take the first logarithm:

$$\begin{aligned} \ln(M(r)) &= \ln\left(\frac{1}{\alpha} \cdot e^{r^{1/\alpha}} \cdot (1 + \epsilon(r))\right) \\ &= \ln\left(\frac{1}{\alpha}\right) + \ln\left(e^{r^{1/\alpha}}\right) + \ln(1 + \epsilon(r)) \\ &= -\ln(\alpha) + r^{1/\alpha} + \ln(1 + \epsilon(r)) \end{aligned}$$

By factoring out the term which grows the fastest, we can write:

$$\ln M(r) = -\ln(\alpha) + r^{1/\alpha} + \ln(1 + \epsilon(r)) = r^{1/\alpha} \left(1 - \frac{\ln \alpha}{r^{1/\alpha}} + \frac{\ln(1 + \epsilon(r))}{r^{1/\alpha}}\right)$$

By taking the second logarithm we obtain:

$$\begin{aligned} \ln(\ln M(r)) &= \ln\left[r^{1/\alpha} \left(1 - \frac{\ln \alpha}{r^{1/\alpha}} + \frac{\ln(1 + \epsilon(r))}{r^{1/\alpha}}\right)\right] \\ &= \ln(r^{1/\alpha}) + \ln\left(1 - \frac{\ln \alpha}{r^{1/\alpha}} + \frac{\ln(1 + \epsilon(r))}{r^{1/\alpha}}\right) \\ &= \frac{1}{\alpha} \ln r + \ln\left(1 - \frac{\ln \alpha}{r^{1/\alpha}} + \frac{\ln(1 + \epsilon(r))}{r^{1/\alpha}}\right) \end{aligned}$$

If we substitute the values we obtained into the limit supremum for growth degree:

$$\rho = \limsup_{r \rightarrow \infty} \frac{\frac{1}{\alpha} \ln r + \ln \left( 1 - \frac{\ln \alpha}{r^{1/\alpha}} + \frac{\ln(1+\epsilon(r))}{r^{1/\alpha}} \right)}{\ln r}$$

Let's divide the numerator by  $\ln r$ :

$$\rho = \limsup_{r \rightarrow \infty} \left[ \frac{1}{\alpha} + \frac{\ln \left( 1 - \frac{\ln \alpha}{r^{1/\alpha}} + \frac{\ln(1+\epsilon(r))}{r^{1/\alpha}} \right)}{\ln r} \right]$$

Let's examine the terms in the numerator when  $r \rightarrow \infty$ :

$$\lim_{r \rightarrow \infty} \frac{\ln \alpha}{r^{1/\alpha}} = 0, \quad \lim_{r \rightarrow \infty} \frac{\ln(1+\epsilon(r))}{r^{1/\alpha}} = 0$$

Because that the expressions given above approaches infinity as  $r \rightarrow \infty$ , logarithm in the numerator approaches  $\ln 1$  which is equal to 0. By substituting these results, we get a formula for the growth degree with a great clarity:

$$\begin{aligned} \rho &= \limsup_{r \rightarrow \infty} \left[ \frac{1}{\alpha} + \frac{\ln \left( 1 - \frac{\ln \alpha}{r^{1/\alpha}} + \frac{\ln(1+\epsilon(r))}{r^{1/\alpha}} \right)}{\ln r} \right] = \lim_{r \rightarrow \infty} \left[ \frac{1}{\alpha} + \frac{\ln(1-0+0)}{\ln r} \right] \\ &= \lim_{r \rightarrow \infty} \left( \frac{1}{\alpha} + \frac{\ln(1)}{\ln r} \right) = \boxed{\frac{1}{\alpha} + \frac{0}{\infty} = \frac{1}{\alpha}}. \end{aligned}$$